Status	Finished
Started	Sunday, 18 October 2026, 10:16 AM
Completed	Sunday, 18 October 2026, 10:25 AM
Duration	8 mins 53 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Deleting data from a list
b. Inserting data into a list
c. Arranging data in a particular format
d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **2**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer
b. Greedy method
c. Backtracking
d. Dynamic programming

The correct answer is:
Divide and conquer

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Arranging data in a specific order
- c. Removing duplicates from a list
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Decreases the efficiency of algorithms
- c. Makes data harder to manage
- d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It compares adjacent elements and swaps them if necessary
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **6**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It uses the heap data structure
c. It bubbles up the largest element in each pass
d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting helps in finding duplicates quickly
b. Sorting is rarely used in programming
c. Sorting decreases the efficiency of selection operations
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **8**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting
b. Complexity
c. Searching
d. Merging

The correct answer is:
Complexity

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort
b. Merge sort
c. Selection sort
d. Heap sort

The correct answer is:
Merge sort

Question **10**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120
b. 40,70,80,90,120,150,200,300,400,470
c. 80,150,200,470,40,90,300,400,70,120
d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **11**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. order()
b. arrange()
c. sorted()
d. sort()

The correct answer is: sorted()

Question **12**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of all keys

b. A sorting algorithm is stable if it preserves the order of duplicate keys

c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort

b. Bubble Sort

c. Binary Search

d. Linear Search

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting

b. Sorting data sequentially

c. Dividing the input into parts, solving each part, and combining the solutions

d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **15**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Sorting the entire list in one pass
b. Bubbling up the smallest element
c. Dividing the list into halves
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position